CS 6515/4540 (Fall 2026)

Exam 2 - PRACTICE

Name: _____

GTID: _____

GT Username: _____

INSTRUCTIONS

1. When the exam starts, write your name (or GT ID or GT Username) on each page.
2. Bringing two pages of notes is permitted.
3. No books, calculators, or other electronic devices permitted.
4. Make your answers clear and concise.
5. ONLY write on the **FRONT** of each sheet. Backs will not be scanned.
6. You may refer to standard algorithms from class without giving full pseudocode.

I am aware of and in accordance with the Academic Honor Code of Georgia Tech and the Georgia Tech Code of Conduct. I will use no person's help on this test. Also, I have read all the instructions on this page.

Signature:

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1 Short Answer Questions (36 points, each part 6 points)

Unless explicitly mentioned, a short 1–2 line justification is required.

1. **LP.** Every convex set can be written as an intersection of a finite number of half-spaces (linear inequalities).

   ◯ True

   ◯ False

2. **Facility Location.**

   Consider the LP relaxation for the facility location problem from class, where I is the set of potential facility locations and J is the set of clients such that $I \cap J = \emptyset$, i.e., no client share a location with a potential facility:

   $$\min \sum_{i \in I} f_i y_i + \sum_{i \in I} \sum_{j \in J} d_{ij} x_{ij}, \quad \text{s.t. } \sum_{i \in I} x_{ij} = 1 \ \forall j \in J, \quad x_{ij} \leq y_i \ \forall i, j, \quad x_{ij}, y_i \geq 0.$$

   Suppose all opening costs are zero, i.e., $f_i = 0$ for all i. What happens to the optimal LP value?

   (A) The optimal LP value is 0.

   (B) Every client connects to its nearest facility.

   (C) The LP becomes unbounded.

   (D) The LP value equals the total connection cost of the MST.

3. **Farkas' Lemma.** Let $A = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$, $b = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$. Use Farkas to show $Ax = b, x \geq 0$ is infeasible by giving a suitable λ_1, λ_2.

4. **Gradient Descent** When running gradient descent on a β-smooth, convex function, we will converge to the optimum regardless of step size or initial point chosen

   ◯ True

   ◯ False

5. **Convex Bodies** We can run Gradient Descent to solve LPs in polynomial time, assuming the feasible region of the LP is a **bounded** convex set and the LP objective is a linear function, which is convex.

   ◯ True

   ◯ False

6. **Approximation Algorithms**

A 2-approximation algorithm is applied to a maximization problem, and the optimal solution is known to be 5. Which of these are possible values the approximation algorithm may return? **Choose all that apply.**

A. 2.5

B. 5

C. 7

D. 10

E. None of the above.

2 Linear Programming (25 points)

1. **Polygon (10 points).** Let P be a convex polygon given by

$$a_i x + b_i y \leq c_i \quad (i = 1, \ldots, m).$$

Given a fixed point $q = (x_0, y_0)$, find $p = (x_1, y_1) \in P$ minimizing

$$\|p - q\|_\infty = \max\{|x_1 - x_0|,\ |y_1 - y_0|\}.$$

Formulate a Linear Program that computes the minimum L_∞ distance from q to P. (*Hint:* Introduce t with $|x_1 - x_0| \leq t$, $|y_1 - y_0| \leq t$ and linearize.)

2. **Matching.** Consider the (non-bipartite) maximum matching LP relaxation on $G = (V, E)$ with $\delta(v)$ denoting all the edges incident to any vertex $v \in V$:

$$\max \sum_{e \in E} x_e \quad \text{s.t.} \quad \sum_{e \in \delta(v)} x_e \leq 1 \ \forall v \in V, \qquad x_e \geq 0 \ \forall e \in E.$$

(a) LP Relaxation (5 points) Prove that for any graph $G = (V, E)$, the optimal value of this LP relaxation is at least the size of the maximum matching in G.

(b) Integrality gap (5 points) Give an example of a graph G where the LP relaxation value is strictly more than the size of the maximum matching on G.

(*Hint:* There is a graph already on just 3 vertices.)

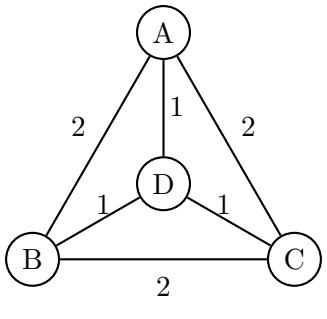
(c) Dual of the LP (5 points). Write the dual of the above max-matching LP relaxation.

3 Steiner Tree (19 Points)

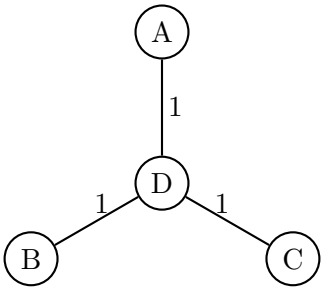
Suppose we are given a fully-connected (complete) graph $G = (V, E)$ with nonnegative edge weights satisfying the triangle inequality, and a subset $T \subseteq V$ of vertices that must be connected. Find a minimum-weight subset of edges $F \subseteq E$ that connects all vertices in T (vertices in $V \setminus T$ may or may not be connected).
(*Note:* The optimal solution is a tree because otherwise we can drop a cycle edge while maintaining connectivity. Finding the minimum such tree (known as Steiner tree) is NP-complete.)

Let G_T be a fully-connected subgraph of G induced by T (i.e., the subgraph where we remove vertices $V \setminus T$). In this problem we will prove that the minimum spanning tree (MST) of G_T is a 2-approximation to the minimum

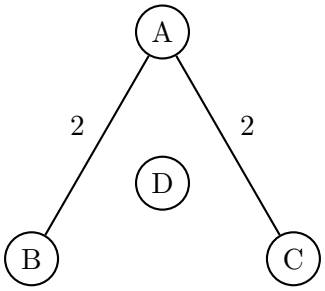
Example.



Graph with $T = \{A, B, C\}$



Steiner Tree



MST of G_T

1. (12 points) Let F^* denote the subset of edges that form the minimum Steiner tree. Prove how we can use it to form a spanning tree of G_T (note this spanning tree cannot visit any vertex in $V \setminus T$ since G_T only has vertices T) of total weight at most twice the total weight of F^*.

   (*Hint:* The proof is similar to why MST is a 2-approx for TSP. Recall that G is complete and edge-weights satisfy triangle inequality.)

2. (7 points) Using part 1 (even if you have not proved it), deduce that MST of G_T is a 2-approx for minimum Steiner tree.

4 Convex Optimization (20 points)

Definition 1 (strong convexity)**.** *A differentiable f is α-strongly convex, for $\alpha > 0$, if for all $x, y \in \mathbb{R}^d$,*

$$\boxed{f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{\alpha}{2} \|y - x\|^2}$$

Consider an α-strongly convex and β-smooth function $f : \mathbb{R}^d \to \mathbb{R}$ with a global minimum at some $x^* \in \mathbb{R}^d$. We run gradient descent with no constraints (no projection) on f beginning from some point x_0, with step size $\eta = \frac{1}{\beta}$. Let $x_k \in \mathbb{R}^d$ be the iterate after k steps:

$$x_{k+1} = x_k - \frac{1}{\beta} \nabla f(x_k).$$

1. (8 points) Using the definition of β-smoothness and step size $\eta = 1/\beta$, show that

$$\frac{1}{2\beta} \|\nabla f(x_k)\|^2 \ \leq \ f(x_k) - f(x_{k+1}).$$

2. (6 points) For any α-strongly convex function f with global minimum at x^*, it can be shown that

$$\|\nabla f(x_k)\|^2 \ \geq \ 2\alpha \big(f(x_k) - f(x^*)\big).$$

Assuming this and part 1, prove that

$$f(x_{k+1}) - f(x^*) \ \leq \ \left(1 - \frac{\alpha}{\beta}\right) \big(f(x_k) - f(x^*)\big).$$

3. (6 points) Using the results above show that for any $k \geq 0$

$$f(x_k) - f(x^*) \leq \left(1 - \frac{\alpha}{\beta}\right)^k (f(x_0) - f(x^*)),$$

i.e., gradient descent on α-strongly convex and β-smooth functions converges exponentially fast.

END OF EXAM